Breaking the Spurious Causality of Conditional Generation via Fairness Intervention with Corrective Sampling

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Bias in Generative Model

Generative model inherits the bias in the training dataset

- BigGAN trained on CelebA dataset generate fewer minority group samples than majority group samples



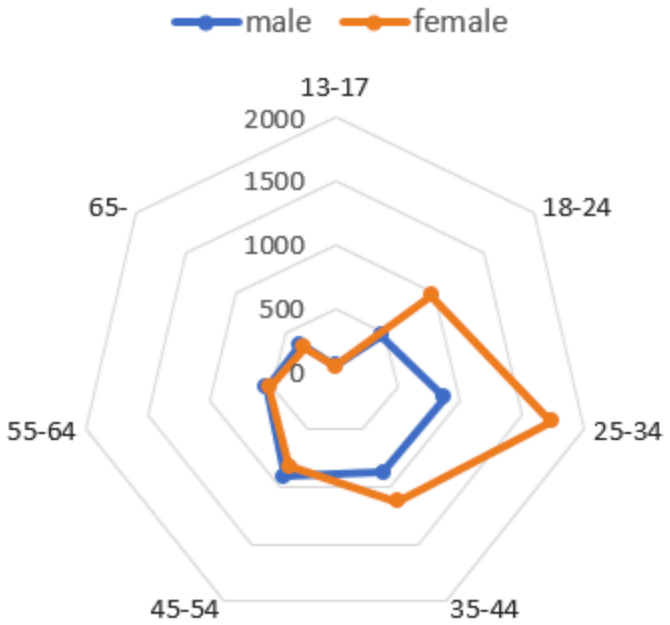
[Choi et al., 2020]

[Choi et al., 2020] Fair Generative Modeling via Weak Supervision, ICML 2020

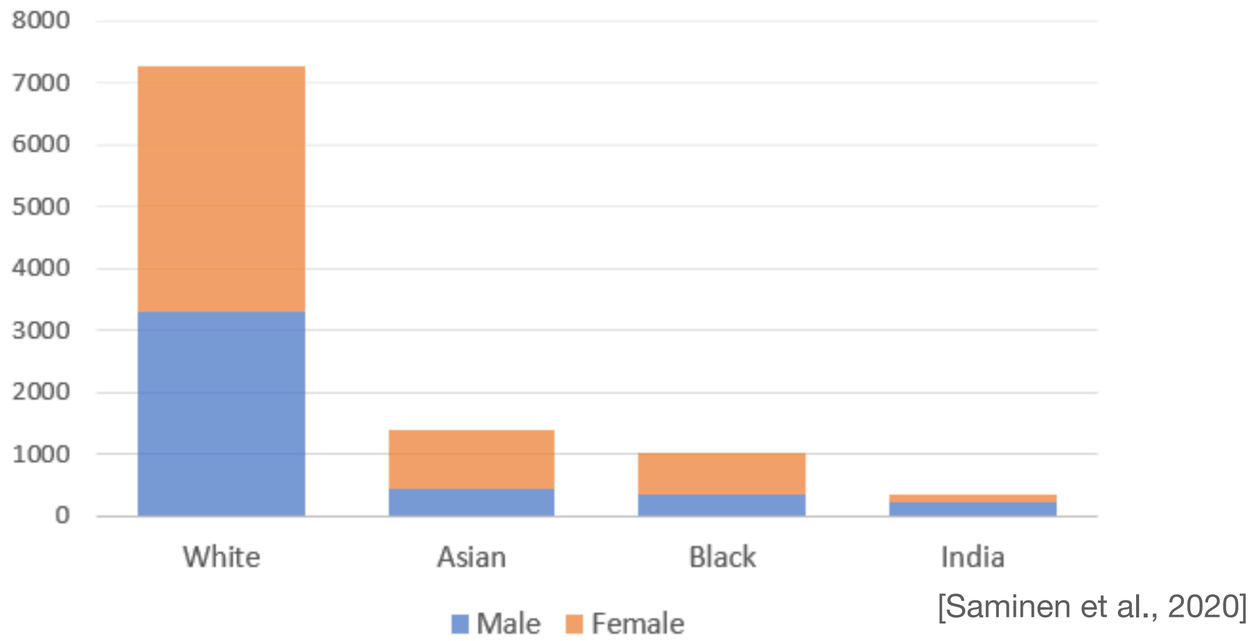
Bias in Generative Model

Generative model inherits the bias in the training dataset

- StyleGAN trained on CelebA-HQ and FFHQ datasets generate fewer minority group samples than majority group samples



Gender counts of the generated pictures by age



[Saminen et al., 2020]

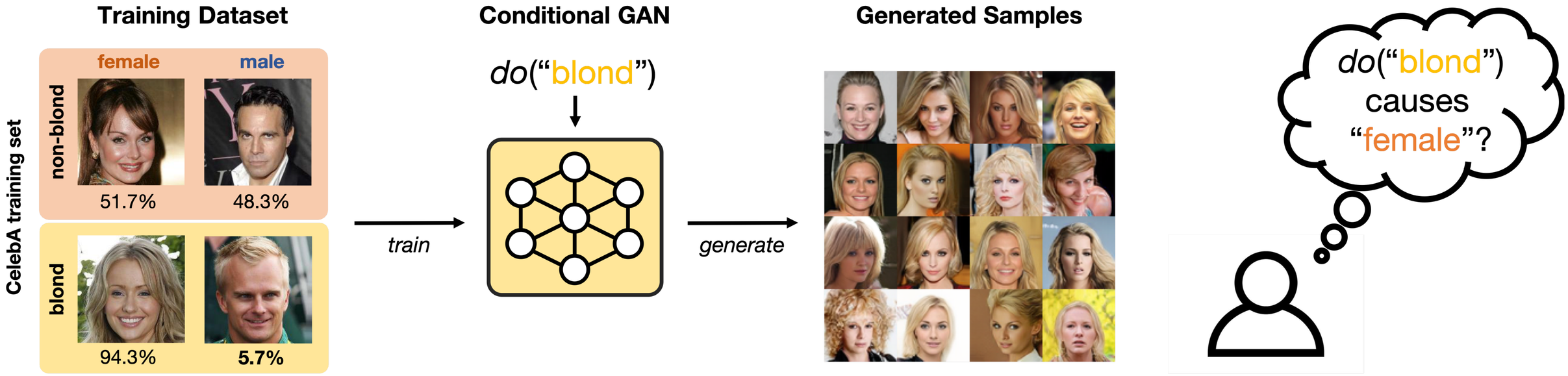
Counts of generated images by race

[Salminen et al., 2020] Analyzing demographic bias in artificially generated facial pictures, CHI 2020

Bias in Conditional Generative Model

Conditional generation faces more problematic fairness concern

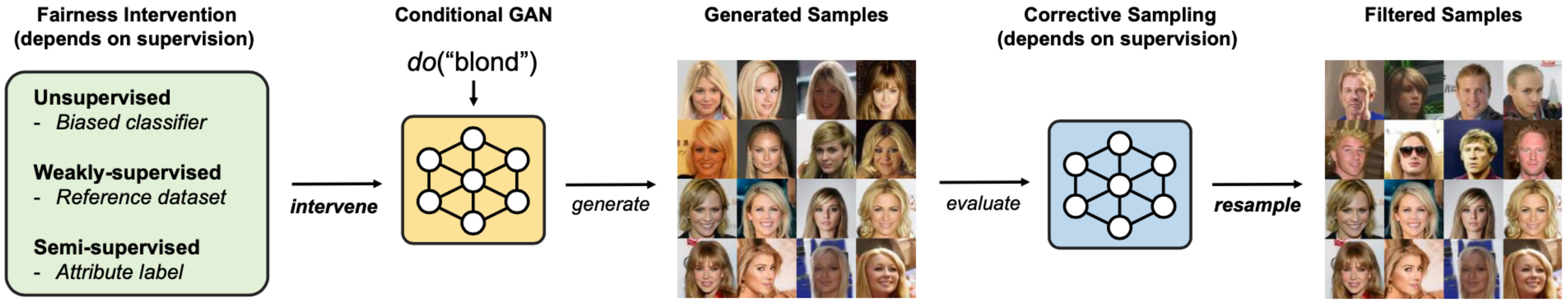
- There is causal relationship between act of conditioning and the distribution of generated samples, which can amplify bias beliefs held by users



"Spurious Causality of Conditional Generation"

Fairness Intervention with Corrective Sampling

A general strategy for mitigate the spurious causality of conditional generation

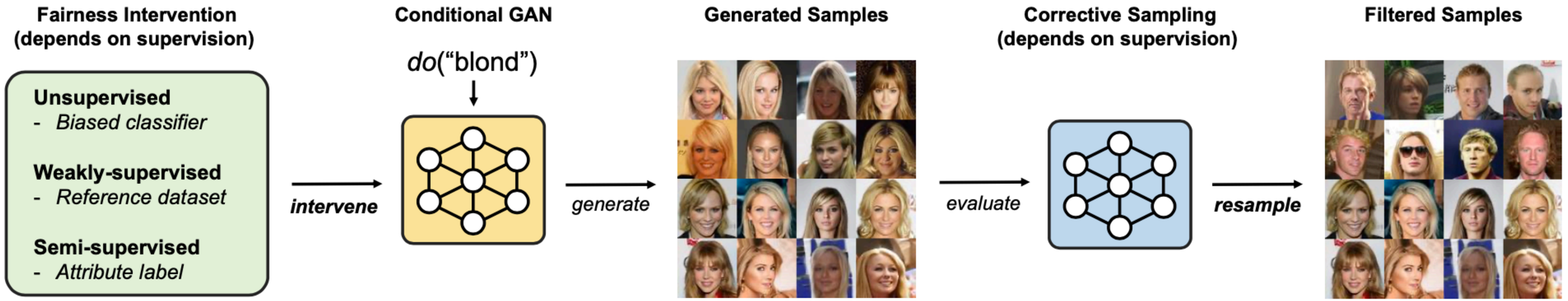


Fairness Intervention (FI)

- Encourage generator to synthesize minority samples during the training phase
- The specific intervention rule depends on the degree of supervision available on the latent attributes

Fairness Intervention with Corrective Sampling

A general strategy for mitigate the spurious causality of conditional generation



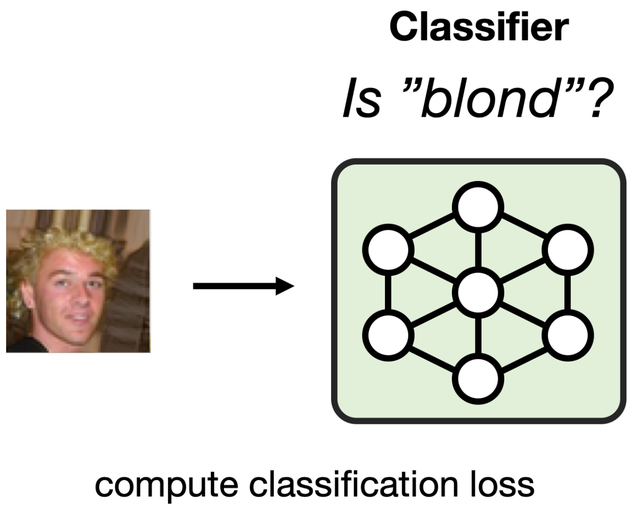
Corrective Sampling (CS)

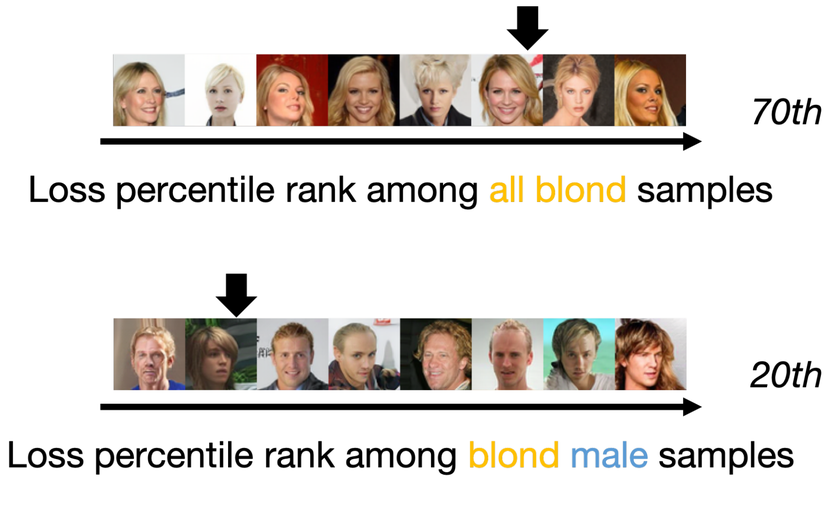
- Apply rejection sampling on the synthesized samples during the generation phase
- The criteria for rejection depend on the desired goal

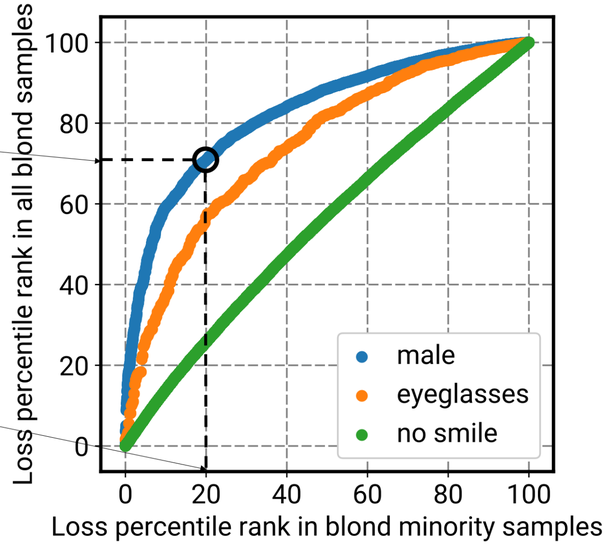
FICS - Unsupervised

Learning the data distribution

Unsupervised: Utilize a correlation between the misprediction of the biased classifier and the generation frequency of the biased cGAN



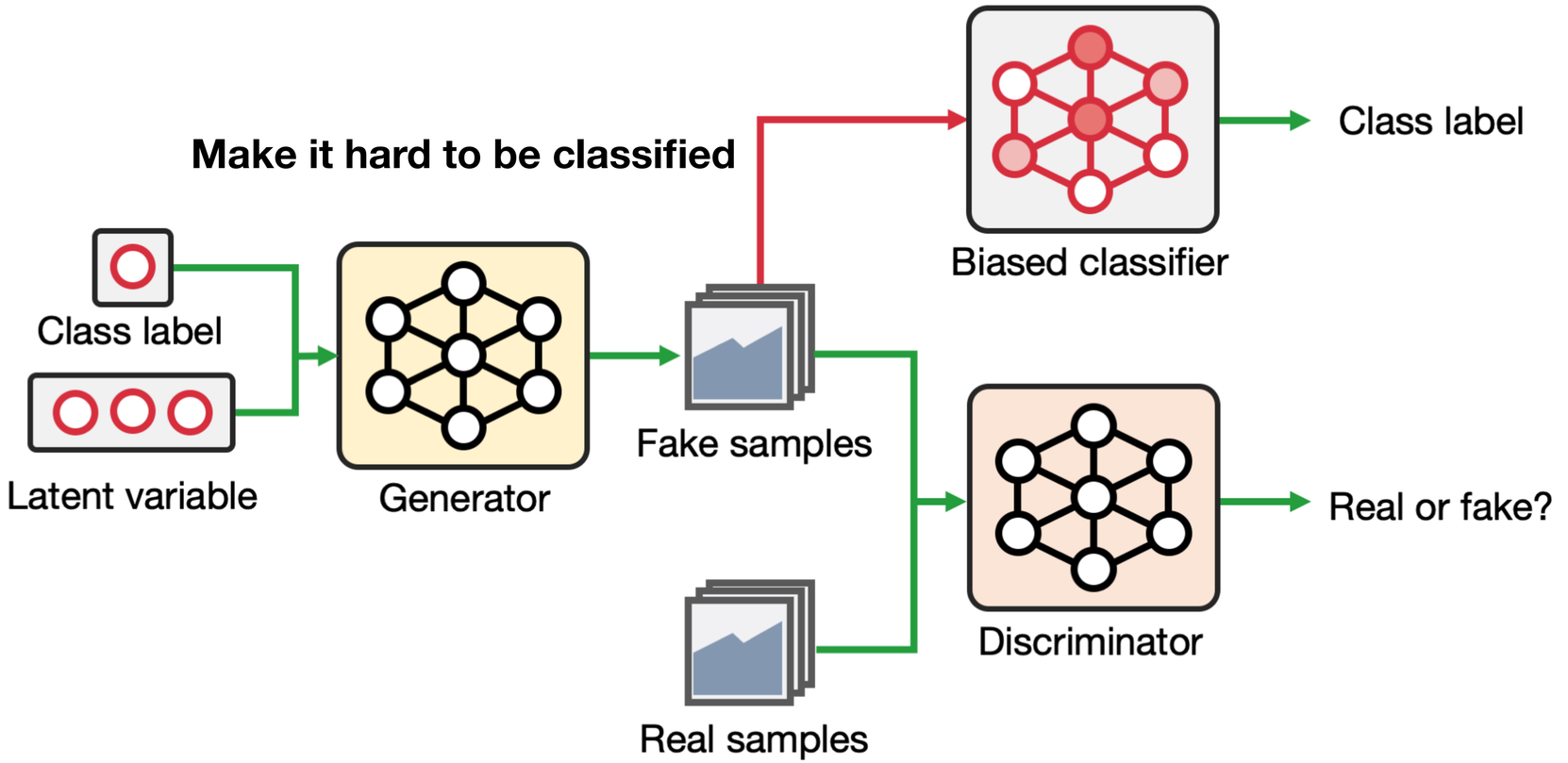




FICS - Unsupervised

Learning the data distribution

FI step - encourage generator to synthesize more samples that are likely to be mispredicted by a biased classifier



$$\sum_{y' \neq y} \mathrm{CE}(f_b(G(z, y)), y')$$

Minimize classification loss over the wrong targets

$$\mathcal{L}_{\mathrm{cGAN}}$$

FICS - Unsupervised

Learning the data distribution

FI step - encourage generator to synthesize more samples that are likely to be mispredicted by a biased classifier

$$\min_G \max_D \mathcal{L}_{\text{cGAN}} + \lambda \cdot \sum_{y' \neq y} \text{CE}(f_b(G(z, y)), y')$$

CS step - Discriminator rejection sampling to recover the data distribution

- The regularizer used in FI step can distort the training distribution

FICS - Semi-supervised

Learning the fair distribution

Semi-supervised: Small number of samples with sensitive attribute labels

- Train an attribute classifier f_a to estimate the sensitive attribute of unlabeled samples

FI step - Data resampling inversely proportional to the group population

$$w(x) = \frac{1}{|\{(x', y', a') \in \mathcal{D} : (y', a') = (y, f_a(x))\}|}$$

CS step - Rejection sampling to control the desired group ratio $p_{\text{fair}}(a|y)$

- Use the attribute classifier f_a to explicitly control the group ratio of generated samples

$$p_{\text{fair}}(x|y) = \sum_a p_{\text{data}}(x|a, y) p_{\text{fair}}(a|y)$$

Experiments

Unsupervised: Learning the data distribution

	Intra-FID (↓)		Minor in Blond (%)		
	Blond	Non-blond	Male	Eyeglasses	No lipstick
Dataset			5.72	1.66	18.78
Estimated			5.85	1.85	17.61
Unsupervised					
cGAN (Mirza & Osindero, 2014)	1.86	1.64	3.97	0.88	11.04
Dia-GAN (Lee et al., 2021)	1.88	1.84	5.06	1.49	14.10
FICS (ours)	**1.78**	**1.38**	**6.13**	**1.90**	**16.34**

Experiments

Supervised: Learning the fair distribution

	Target Male ratio: 30%		Target Male ratio: 50%	
	Fair Intra-FID (↓) Blond	Occurrence of Male in Blond (%)	Fair Intra-FID (↓) Blond	Occurrence of Male in Blond (%)
Reference		30.00		50.00
Estimated		28.60		47.15
Weakly-supervised				
Choi et al. (2020)	10.71	7.60	10.70	18.34
FICS (ours)	**10.36**	**15.65**	**10.24**	**27.04**
Semi-supervised				
FairGen (Tan et al., 2020)	12.71	**29.54**	11.98	49.76
FICS (ours)	**10.56**	25.72	**11.68**	**47.11**

Conclusion

We raise the issue of **spurious causality of conditional generation**,
which occurs from conditional generative models trained on the dataset with spurious correlation

We propose **Fairness Intervention with Corrective Sampling**,
an effective remedy for the spurious causality of conditional generation.

Our proposed method works on various scenarios with different levels of supervision on spurious attributes

Thank you for your attention!